

Practice Final Exam Solutions (STAT 461)

Problem 1

Question. The world population has recently crossed the 7.5 billion mark, but approximately 1 in 8 people in the world today remain undernourished or malnourished. If 120 people are randomly selected, find the chances that there are (a) at least 5, (b) at most 9, and (c) between 10 and 20 inclusive undernourished or malnourished people in the sample.

Solution. Let

X = number of undernourished or malnourished people in the sample.

Since approximately 1 in 8 people are undernourished or malnourished,

$$p = \frac{1}{8} = 0.125, \quad n = 120.$$

Thus

$$X \sim \text{Binomial}(120, 0.125).$$

Since

$$np = 120(0.125) = 15, \quad n(1 - p) = 120(0.875) = 105,$$

the binomial distribution may be approximated by a normal distribution:

$$X \approx N(np, np(1 - p)) = N(15, 13.125).$$

Therefore,

$$\mu = 15, \quad \sigma = \sqrt{13.125} \approx 3.6228.$$

(a) Using a continuity correction,

$$P(X \geq 5) = P(X > 4.5).$$

Thus,

$$P(X \geq 5) \approx P\left(Z > \frac{4.5 - 15}{3.6228}\right) = P(Z > -2.899).$$

Therefore,

$$\boxed{P(X \geq 5) \approx 0.9981}.$$

(b) Using a continuity correction,

$$P(X \leq 9) = P(X < 9.5).$$

Thus,

$$P(X \leq 9) \approx P\left(Z < \frac{9.5 - 15}{3.6228}\right) = P(Z < -1.518).$$

Therefore,

$$\boxed{P(X \leq 9) \approx 0.0645}.$$

(c) Using a continuity correction,

$$P(10 \leq X \leq 20) = P(9.5 < X < 20.5).$$

Thus,

$$P(10 \leq X \leq 20) \approx P\left(\frac{9.5 - 15}{3.6228} < Z < \frac{20.5 - 15}{3.6228}\right).$$

That is,

$$P(10 \leq X \leq 20) \approx P(-1.518 < Z < 1.518).$$

Therefore,

$$\boxed{P(10 \leq X \leq 20) \approx 0.8710}.$$

Problem 2

Question. If X , Y , and Z are independent random variables with means 15, 22, and 10 respectively and standard deviations 2, 5, and 7 respectively, find the mean and the standard deviation of $3X + 2Y - 4Z$.

Solution. Let

$$W = 3X + 2Y - 4Z.$$

Using linearity of expectation,

$$E(W) = 3E(X) + 2E(Y) - 4E(Z).$$

Substituting the given means,

$$E(W) = 3(15) + 2(22) - 4(10) = 45 + 44 - 40 = 49.$$

Thus,

$$\boxed{\mu_W = 49}.$$

Since X, Y, Z are independent,

$$\text{Var}(W) = 3^2 \text{Var}(X) + 2^2 \text{Var}(Y) + (-4)^2 \text{Var}(Z).$$

Using the given standard deviations,

$$\text{Var}(X) = 2^2 = 4, \quad \text{Var}(Y) = 5^2 = 25, \quad \text{Var}(Z) = 7^2 = 49.$$

Therefore,

$$\text{Var}(W) = 9(4) + 4(25) + 16(49) = 36 + 100 + 784 = 920.$$

So,

$$\boxed{\sigma_W = \sqrt{920} = 2\sqrt{230} \approx 30.33}.$$

Problem 3

Question. If $X_1, \dots, X_n$ are i.i.d. Standard Beta(3, 5) random variables, find the mean and variance of

$$X = \sum_{i=1}^n (c_i X_i + b_i),$$

for constants $c_1, \dots, c_n$ and $b_1, \dots, b_n$.

Solution. For a Beta(α, β) random variable,

$$E(X_i) = \frac{\alpha}{\alpha + \beta}, \quad \text{Var}(X_i) = \frac{\alpha\beta}{(\alpha + \beta)^2(\alpha + \beta + 1)}.$$

Here, $\alpha = 3$ and $\beta = 5$. Thus,

$$E(X_i) = \frac{3}{3 + 5} = \frac{3}{8},$$

and

$$\text{Var}(X_i) = \frac{(3)(5)}{(3 + 5)^2(3 + 5 + 1)} = \frac{15}{64(9)} = \frac{15}{576} = \frac{5}{192}.$$

Now,

$$E(X) = E \left[\sum_{i=1}^n (c_i X_i + b_i) \right].$$

By linearity of expectation,

$$E(X) = \sum_{i=1}^n (c_i E(X_i) + b_i).$$

Therefore,

$$E(X) = \sum_{i=1}^n \left(\frac{3c_i}{8} + b_i \right).$$

Since $X_1, \dots, X_n$ are independent,

$$\text{Var}(X) = \text{Var} \left[\sum_{i=1}^n (c_i X_i + b_i) \right] = \sum_{i=1}^n c_i^2 \text{Var}(X_i).$$

Thus,

$$\text{Var}(X) = \frac{5}{192} \sum_{i=1}^n c_i^2.$$

Problem 4

Question. Durations of 911 phone calls are skewed to the right, with population mean 12 minutes and population standard deviation 3.2 minutes. For a simple random sample of 64 calls, find the approximate distribution of the sample average. Then find the chances that the sample average (a) exceeds 13 minutes and (b) is below 10.8 minutes.

Solution. Let X denote the duration of a randomly selected 911 phone call. The given values are

$$\mu = 12, \quad \sigma = 3.2, \quad n = 64.$$

Although the population distribution is skewed to the right, the sample size is large enough to use the Central Limit Theorem. Thus,

$$\bar{X} \approx N\left(\mu, \frac{\sigma^2}{n}\right).$$

Therefore,

$$\bar{X} \approx N\left(12, \frac{3.2^2}{64}\right) = N(12, 0.16).$$

Equivalently,

$$\boxed{\bar{X} \approx N(12, 0.4^2)}.$$

(a)

$$P(\bar{X} > 13) = P\left(Z > \frac{13 - 12}{0.4}\right) = P(Z > 2.5).$$

Thus,

$$P(\bar{X} > 13) = 1 - \Phi(2.5) \approx 1 - 0.9938 = 0.0062.$$

Therefore,

$$\boxed{P(\bar{X} > 13) \approx 0.0062}.$$

(b)

$$P(\bar{X} < 10.8) = P\left(Z < \frac{10.8 - 12}{0.4}\right) = P(Z < -3).$$

Thus,

$$P(\bar{X} < 10.8) = \Phi(-3) \approx 0.0013.$$

Therefore,

$$\boxed{P(\bar{X} < 10.8) \approx 0.0013}.$$

Problem 5

Question. Let $Y_1, \dots, Y_{36}$ be a simple random sample from a Poisson($\mu = 9$) population. What is the approximate distribution of $2(\bar{Y} - 9)$? So, what is $P(|\bar{Y} - 9| \leq 1.5)$?

Solution. For a Poisson random variable,

$$E(Y_i) = \mu = 9, \quad \text{Var}(Y_i) = \mu = 9.$$

The sample mean is

$$\bar{Y} = \frac{1}{36} \sum_{i=1}^{36} Y_i.$$

By the Central Limit Theorem,

$$\bar{Y} \approx N\left(\mu, \frac{\sigma^2}{n}\right) = N\left(9, \frac{9}{36}\right) = N\left(9, \frac{1}{4}\right).$$

Thus,

$$\bar{Y} - 9 \approx N\left(0, \frac{1}{4}\right).$$

Therefore,

$$2(\bar{Y} - 9) \approx N(0, 1).$$

So,

$$\boxed{2(\bar{Y} - 9) \approx N(0, 1)}.$$

Now,

$$P(|\bar{Y} - 9| \leq 1.5) = P(-1.5 \leq \bar{Y} - 9 \leq 1.5).$$

Multiplying all parts by 2,

$$P(|\bar{Y} - 9| \leq 1.5) = P(-3 \leq 2(\bar{Y} - 9) \leq 3).$$

Since $2(\bar{Y} - 9) \approx Z$,

$$P(|\bar{Y} - 9| \leq 1.5) \approx P(-3 \leq Z \leq 3).$$

Using the standard normal table,

$$P(-3 \leq Z \leq 3) = \Phi(3) - \Phi(-3) \approx 0.9987 - 0.0013 = 0.9974.$$

Hence,

$$\boxed{P(|\bar{Y} - 9| \leq 1.5) \approx 0.9974}.$$

Problem 6

Question. In a poll of $n = 1100$ adults, 517 said they favored legalizing marijuana. Write down a 90% confidence interval and a 99% confidence interval for the true population proportion supporting legalization.

Solution. Let p be the true unknown population proportion of adults who support legalizing marijuana. The given values are

$$n = 1100, \quad x = 517.$$

Thus,

$$\hat{p} = \frac{x}{n} = \frac{517}{1100} = 0.47.$$

Also,

$$1 - \hat{p} = 0.53.$$

For a large-sample confidence interval for a population proportion,

$$\hat{p} \pm z_{\alpha/2} \sqrt{\frac{\hat{p}(1 - \hat{p})}{n}}.$$

First compute the standard error:

$$SE = \sqrt{\frac{\hat{p}(1 - \hat{p})}{n}} = \sqrt{\frac{(0.47)(0.53)}{1100}} \approx 0.01505.$$

90% Confidence Interval. For a 90% confidence interval,

$$z_{\alpha/2} = z_{0.05} = 1.645.$$

Therefore,

$$0.47 \pm 1.645(0.01505) = 0.47 \pm 0.02475.$$

Thus,

$$\boxed{(0.4453, 0.4947)}.$$

99% Confidence Interval. For a 99% confidence interval,

$$z_{\alpha/2} = z_{0.005} = 2.576.$$

Therefore,

$$0.47 \pm 2.576(0.01505) = 0.47 \pm 0.03877.$$

Thus,

$$\boxed{(0.4312, 0.5088)}.$$

Problem 7

Question. How large a sample size is needed to estimate an unknown population proportion of “yes” answers by a 99% confidence interval whose width is 0.4?

Solution. For a confidence interval for a population proportion,

$$\hat{p} \pm z_{\alpha/2} \sqrt{\frac{\hat{p}(1 - \hat{p})}{n}}.$$

The width of the interval is $2E$, where E is the margin of error. The desired width is 0.4, so

$$2E = 0.4 \quad \implies \quad E = 0.2.$$

Since the population proportion is unknown, use the conservative value

$$\hat{p} = 0.5.$$

Thus,

$$\hat{p}(1 - \hat{p}) = (0.5)(0.5) = 0.25.$$

For a 99% confidence interval,

$$z_{\alpha/2} = z_{0.005} = 2.576.$$

The required sample size satisfies

$$E = z_{\alpha/2} \sqrt{\frac{\hat{p}(1 - \hat{p})}{n}}.$$

Solving for n ,

$$n = \frac{z_{\alpha/2}^2 \hat{p}(1 - \hat{p})}{E^2}.$$

Substituting,

$$n = \frac{(2.576)^2(0.25)}{(0.2)^2} = \frac{6.6358(0.25)}{0.04} = 41.474.$$

Since the sample size must be rounded up,

$$\boxed{n = 42}.$$

Problem 8

Question. If a random sample of size 15 is taken from a Normal population with mean 75, what are the chances that

$$\left(\frac{\bar{X} - 75}{S/\sqrt{15}} \right)$$

exceeds 2.145 in absolute value?

Solution. Since the population standard deviation is not given, use the sample standard deviation S . Then

$$T = \frac{\bar{X} - \mu}{S/\sqrt{n}}$$

has a Student's t -distribution with

$$n - 1 = 15 - 1 = 14$$

degrees of freedom. Here,

$$T = \frac{\bar{X} - 75}{S/\sqrt{15}} \sim t_{14}.$$

Find

$$P \left(\left| \frac{\bar{X} - 75}{S/\sqrt{15}} \right| > 2.145 \right) = P(|T| > 2.145).$$

From the t -table with 14 degrees of freedom,

$$t_{0.025,14} = 2.145.$$

Therefore,

$$P(T > 2.145) = 0.025.$$

By symmetry of the t -distribution,

$$P(|T| > 2.145) = 2P(T > 2.145) = 2(0.025) = 0.05.$$

Hence,

$$\boxed{P \left(\left| \frac{\bar{X} - 75}{S/\sqrt{15}} \right| > 2.145 \right) = 0.05}.$$

Problem 9

Question. Suppose $Y_1, \dots, Y_n$ are a simple random sample from a Uniform $[0, \theta]$ population. What is the method-of-moments estimate of θ ? What is the method-of-moments estimate of θ^2 ?

Solution. For a Uniform $[0, \theta]$ random variable,

$$E(Y_i) = \frac{0 + \theta}{2} = \frac{\theta}{2}.$$

The first moment equation is

$$\bar{Y} = \frac{\theta}{2}.$$

Solving for θ ,

$$\boxed{\hat{\theta}_{\text{MOM}} = 2\bar{Y}}.$$

For a Uniform $[0, \theta]$ random variable,

$$\text{Var}(Y_i) = \frac{(\theta - 0)^2}{12} = \frac{\theta^2}{12}.$$

Equating this to the sample variance,

$$S^2 = \frac{1}{n-1} \sum_{i=1}^n (Y_i - \bar{Y})^2,$$

gives

$$S^2 = \frac{\theta^2}{12}.$$

Solving for θ^2 ,

$$\boxed{\hat{\theta}_{\text{MOM}}^2 = 12S^2}.$$

Problem 10

Question. If $X_1, X_2, \dots, X_n$ are a simple random sample from a Weibull($3, \beta$) population, derive the maximum likelihood estimate of β .

Solution. The Weibull density is

$$f(x) = \frac{\alpha}{\beta^\alpha} x^{\alpha-1} e^{-(x/\beta)^\alpha}, \quad x > 0.$$

Since $\alpha = 3$,

$$f(x) = \frac{3}{\beta^3} x^2 e^{-(x/\beta)^3}, \quad x > 0.$$

The likelihood function is

$$L(\beta) = \prod_{i=1}^n \frac{3}{\beta^3} x_i^2 e^{-(x_i/\beta)^3}.$$

Thus,

$$L(\beta) = \frac{3^n}{\beta^{3n}} \left(\prod_{i=1}^n x_i^2 \right) \exp \left(- \sum_{i=1}^n \frac{x_i^3}{\beta^3} \right).$$

The log-likelihood function is

$$\ell(\beta) = \ln L(\beta) = n \ln 3 - 3n \ln \beta + 2 \sum_{i=1}^n \ln x_i - \frac{\sum_{i=1}^n x_i^3}{\beta^3}.$$

Differentiate with respect to β :

$$\frac{d\ell}{d\beta} = -\frac{3n}{\beta} + \frac{3 \sum_{i=1}^n x_i^3}{\beta^4}.$$

Set this equal to 0:

$$-\frac{3n}{\beta} + \frac{3 \sum_{i=1}^n x_i^3}{\beta^4} = 0.$$

Multiplying by β^4 ,

$$-3n\beta^3 + 3 \sum_{i=1}^n x_i^3 = 0.$$

Therefore,

$$n\beta^3 = \sum_{i=1}^n x_i^3,$$

so

$$\beta^3 = \frac{1}{n} \sum_{i=1}^n x_i^3.$$

Hence,

$$\hat{\beta}_{\text{MLE}} = \sqrt[3]{\frac{1}{n} \sum_{i=1}^n x_i^3}.$$

To check that this gives a maximum,

$$\frac{d^2\ell}{d\beta^2} = \frac{3n}{\beta^2} - \frac{12\sum_{i=1}^n x_i^3}{\beta^5}.$$

At $\hat{\beta}_{\text{MLE}}$,

$$\sum_{i=1}^n x_i^3 = n\hat{\beta}_{\text{MLE}}^3.$$

Thus,

$$\left. \frac{d^2\ell}{d\beta^2} \right|_{\hat{\beta}_{\text{MLE}}} = \frac{3n}{\hat{\beta}_{\text{MLE}}^2} - \frac{12n\hat{\beta}_{\text{MLE}}^3}{\hat{\beta}_{\text{MLE}}^5} = \frac{3n}{\hat{\beta}_{\text{MLE}}^2} - \frac{12n}{\hat{\beta}_{\text{MLE}}^2} = -\frac{9n}{\hat{\beta}_{\text{MLE}}^2} < 0.$$

Therefore,

$$\hat{\beta}_{\text{MLE}} = \sqrt[3]{\frac{1}{n} \sum_{i=1}^n x_i^3}.$$

Problem 11

Question. A random sample of $n = 16$ pygmy goats from West Africa has sample average height 49 cm and sample standard deviation 6.4 cm. Find a 99% confidence interval for the unknown population mean height.

Solution. The given values are

$$n = 16, \quad \bar{x} = 49, \quad s = 6.4.$$

Since the population standard deviation is unknown and the sample size is small, use a one-sample t confidence interval:

$$\bar{x} \pm t_{\alpha/2, n-1} \frac{s}{\sqrt{n}}.$$

For a 99% confidence interval,

$$\alpha = 0.01, \quad \alpha/2 = 0.005, \quad df = n - 1 = 15.$$

From the t -table,

$$t_{0.005, 15} = 2.947.$$

Thus,

$$49 \pm 2.947 \frac{6.4}{\sqrt{16}}.$$

Since

$$\frac{6.4}{\sqrt{16}} = \frac{6.4}{4} = 1.6,$$

the margin of error is

$$2.947(1.6) = 4.7152.$$

Therefore,

$$49 \pm 4.7152 = (44.2848, 53.7152).$$

Hence,

$$\boxed{(44.2848, 53.7152)}.$$

Problem 12

Question. A sample of 9 second graders who participated in regular sports had manual dexterity scores with mean 32.19 and standard deviation 4.34. An independent sample of 7 second graders who did not participate in regular sports had mean 31.68 and standard deviation 4.56. At the 5% significance level, test whether regular-sports participants have a higher mean score.

Solution. Let

μ_1 = mean score for second graders who participated in regular sports

and

μ_2 = mean score for second graders who did not participate in regular sports.

The hypotheses are

$$H_0 : \mu_1 - \mu_2 \leq 0$$

versus

$$H_a : \mu_1 - \mu_2 > 0.$$

The sample information is

$$n_1 = 9, \quad \bar{x}_1 = 32.19, \quad s_1 = 4.34,$$

and

$$n_2 = 7, \quad \bar{x}_2 = 31.68, \quad s_2 = 4.56.$$

Since the population standard deviations are unknown and not assumed equal, use the two-sample t test statistic

$$t = \frac{\bar{x}_1 - \bar{x}_2 - 0}{\sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}}.$$

Substituting,

$$t = \frac{32.19 - 31.68}{\sqrt{\frac{4.34^2}{9} + \frac{4.56^2}{7}}} = \frac{0.51}{\sqrt{\frac{18.8356}{9} + \frac{20.7936}{7}}}.$$

Thus,

$$t = \frac{0.51}{\sqrt{5.0633}} \approx 0.2266.$$

Using the conservative degrees of freedom,

$$df = \min(n_1 - 1, n_2 - 1) = \min(8, 6) = 6.$$

At the 5% significance level for a right-tailed test,

$$t_{0.05,6} = 1.943.$$

The rejection rule is

$$\text{reject } H_0 \text{ if } t > 1.943.$$

Since

$$0.2266 < 1.943,$$

fail to reject H_0 . Therefore, there is not sufficient evidence to claim that second graders who participate in regular sports have a higher mean manual dexterity score.

Fail to reject H_0 .

Problem 13

Question. In Problem 12, how would the test be done differently if the two unknown population standard deviations were known to be the same?

Solution. If the two unknown population standard deviations are known to be equal, then the test should be done using a pooled two-sample t test.

The hypotheses remain

$$H_0 : \mu_1 - \mu_2 \leq 0$$

versus

$$H_a : \mu_1 - \mu_2 > 0.$$

The sample information is

$$n_1 = 9, \quad \bar{x}_1 = 32.19, \quad s_1 = 4.34,$$

and

$$n_2 = 7, \quad \bar{x}_2 = 31.68, \quad s_2 = 4.56.$$

The pooled sample variance is

$$s_p^2 = \frac{(n_1 - 1)s_1^2 + (n_2 - 1)s_2^2}{n_1 + n_2 - 2}.$$

Substituting,

$$s_p^2 = \frac{(9 - 1)(4.34)^2 + (7 - 1)(4.56)^2}{9 + 7 - 2}.$$

Thus,

$$s_p^2 = \frac{8(18.8356) + 6(20.7936)}{14} = \frac{275.4464}{14} \approx 19.6747.$$

So,

$$s_p = \sqrt{19.6747} \approx 4.4356.$$

The test statistic is

$$t = \frac{\bar{x}_1 - \bar{x}_2 - 0}{s_p \sqrt{\frac{1}{n_1} + \frac{1}{n_2}}}.$$

Substituting,

$$t = \frac{32.19 - 31.68}{4.4356 \sqrt{\frac{1}{9} + \frac{1}{7}}} = \frac{0.51}{4.4356 \sqrt{\frac{16}{63}}} \approx 0.2281.$$

The degrees of freedom are

$$df = n_1 + n_2 - 2 = 9 + 7 - 2 = 14.$$

At the 5% significance level for a right-tailed test,

$$t_{0.05, 14} = 1.761.$$

The rejection rule is

$$\text{reject } H_0 \text{ if } t > 1.761.$$

Since

$$0.2281 < 1.761,$$

fail to reject H_0 .

Fail to reject H_0 .

Problem 14

Question. For each combination of sample size and sample proportion, find the margin of error for a 98% confidence interval: (a) $n = 484$ and 98 yes responses, (b) $n = 1089$ and 655 in favor, (c) $n = 650$ and 540 prefer stricter rules.

Solution. For a confidence interval for a population proportion, the margin of error is

$$E = z_{\alpha/2} \sqrt{\frac{\hat{p}(1 - \hat{p})}{n}}.$$

For a 98% confidence interval,

$$\alpha = 0.02, \quad \alpha/2 = 0.01, \quad z_{\alpha/2} = z_{0.01} = 2.326.$$

(a) Here,

$$n = 484, \quad x = 98.$$

Thus,

$$\hat{p} = \frac{98}{484} \approx 0.2025.$$

The margin of error is

$$E = 2.326 \sqrt{\frac{(0.2025)(1 - 0.2025)}{484}} = 2.326 \sqrt{\frac{(0.2025)(0.7975)}{484}} \approx 0.0425.$$

Therefore,

$$\boxed{E \approx 0.0425}.$$

(b) Here,

$$n = 1089, \quad x = 655.$$

Thus,

$$\hat{p} = \frac{655}{1089} \approx 0.6015.$$

The margin of error is

$$E = 2.326 \sqrt{\frac{(0.6015)(1 - 0.6015)}{1089}} \approx 0.0345.$$

Therefore,

$$\boxed{E \approx 0.0345}.$$

(c) Here,

$$n = 650, \quad x = 540.$$

Thus,

$$\hat{p} = \frac{540}{650} \approx 0.8308.$$

The margin of error is

$$E = 2.326 \sqrt{\frac{(0.8308)(1 - 0.8308)}{650}} \approx 0.0342.$$

Therefore,

$$\boxed{E \approx 0.0342}.$$

Problem 15

Question. At the start of an Ebola epidemic, 0.2% of the population had the virus. If a random sample of 1000 residents is taken, find the chances of finding exactly 5 infected people and at most 3 infected people.

Solution. Let

X = number of Ebola-infected people in the sample.

Since 0.2% of the population had the virus,

$$p = 0.002, \quad n = 1000.$$

Thus,

$$X \sim \text{Binomial}(1000, 0.002).$$

Since n is large and p is small, use a Poisson approximation with

$$\lambda = np = 1000(0.002) = 2.$$

Therefore,

$$X \approx \text{Poisson}(2).$$

Exactly 5 infected people:

$$P(X = 5) = \frac{e^{-2}2^5}{5!} = \frac{e^{-2}(32)}{120} \approx 0.0361.$$

Therefore,

$$\boxed{P(X = 5) \approx 0.0361}.$$

At most 3 infected people:

$$P(X \leq 3) = P(X = 0) + P(X = 1) + P(X = 2) + P(X = 3).$$

Using the Poisson pmf,

$$P(X \leq 3) = e^{-2} \left(\frac{2^0}{0!} + \frac{2^1}{1!} + \frac{2^2}{2!} + \frac{2^3}{3!} \right).$$

Thus,

$$P(X \leq 3) = e^{-2} \left(1 + 2 + 2 + \frac{8}{6} \right) = e^{-2} \left(\frac{19}{3} \right) \approx 0.8571.$$

Therefore,

$$\boxed{P(X \leq 3) \approx 0.8571}.$$

Problem 16

Question. A first bowl has 6 red and 2 blue marbles. A second bowl has 2 red and 2 blue marbles. A marble is transferred from the first bowl to the second. Then a marble is drawn from the second bowl and it is red. Find the conditional probability that the transferred marble was red.

Solution. Let

T_R = the transferred marble was red

and

D_R = the marble drawn from the second bowl was red.

From the first bowl,

$$P(T_R) = \frac{6}{8} = \frac{3}{4}, \quad P(T_B) = \frac{2}{8} = \frac{1}{4}.$$

If a red marble was transferred, then the second bowl has 3 red marbles and 2 blue marbles. Thus,

$$P(D_R | T_R) = \frac{3}{5}.$$

If a blue marble was transferred, then the second bowl has 2 red marbles and 3 blue marbles. Thus,

$$P(D_R | T_B) = \frac{2}{5}.$$

Find $P(T_R | D_R)$. By Bayes' rule,

$$P(T_R | D_R) = \frac{P(D_R | T_R)P(T_R)}{P(D_R | T_R)P(T_R) + P(D_R | T_B)P(T_B)}.$$

Substituting,

$$P(T_R | D_R) = \frac{\left(\frac{3}{5}\right)\left(\frac{3}{4}\right)}{\left(\frac{3}{5}\right)\left(\frac{3}{4}\right) + \left(\frac{2}{5}\right)\left(\frac{1}{4}\right)}.$$

So,

$$P(T_R | D_R) = \frac{\frac{9}{20}}{\frac{9}{20} + \frac{2}{20}} = \frac{\frac{9}{20}}{\frac{11}{20}} = \frac{9}{11}.$$

Therefore,

$$\boxed{P(T_R | D_R) = \frac{9}{11} \approx 0.8182}.$$

Problem 17

Question. For six brands of French wine, X is the number of years old and Y is the selling price per bottle. Find the sample correlation r , the least-squares line, and the predicted average Y for $X = 7$.

Number	1	2	3	4	5	6
X	2	4	3	2	5	4
Y	98	135	162	178	221	199

Solution. First compute the sample means:

$$\bar{x} = \frac{2 + 4 + 3 + 2 + 5 + 4}{6} = \frac{20}{6} = \frac{10}{3},$$

and

$$\bar{y} = \frac{98 + 135 + 162 + 178 + 221 + 199}{6} = \frac{993}{6} = 165.5.$$

Now compute

$$S_{xx} = \sum (x_i - \bar{x})^2, \quad S_{yy} = \sum (y_i - \bar{y})^2, \quad S_{xy} = \sum (x_i - \bar{x})(y_i - \bar{y}).$$

Using the data,

$$S_{xx} = 7.3333, \quad S_{yy} = 9857.5, \quad S_{xy} = 169.$$

The sample correlation is

$$r = \frac{S_{xy}}{\sqrt{S_{xx}S_{yy}}}.$$

Thus,

$$r = \frac{169}{\sqrt{(7.3333)(9857.5)}} \approx 0.6286.$$

Therefore,

$$\boxed{r \approx 0.6286}.$$

The least-squares slope is

$$b_1 = \frac{S_{xy}}{S_{xx}} = \frac{169}{7.3333} \approx 23.0455.$$

The least-squares intercept is

$$b_0 = \bar{y} - b_1\bar{x}.$$

So,

$$b_0 = 165.5 - (23.0455)\left(\frac{10}{3}\right) \approx 88.6818.$$

Therefore, the least-squares line is

$$\boxed{\hat{y} = 88.6818 + 23.0455x}.$$

For $x = 7$,

$$\hat{y} = 88.6818 + 23.0455(7) = 250.$$

Thus,

$$\boxed{\hat{y} = 250}.$$

Problem 18

Question. When rolling a balanced die twice, write down the pmf table for $X =$ the average of the two numbers. Find its mean and standard deviation.

Solution. Let the two die rolls be D_1 and D_2 . Then

$$X = \frac{D_1 + D_2}{2}.$$

There are 36 equally likely outcomes when rolling a balanced die twice. The possible values of $D_1 + D_2$ are $2, 3, \dots, 12$, so the possible values of X are

$$1, \frac{3}{2}, 2, \frac{5}{2}, 3, \frac{7}{2}, 4, \frac{9}{2}, 5, \frac{11}{2}, 6.$$

The pmf table is

x	1	$\frac{3}{2}$	2	$\frac{5}{2}$	3	$\frac{7}{2}$	4	$\frac{9}{2}$	5	$\frac{11}{2}$	6
$P(X = x)$	$\frac{1}{36}$	$\frac{2}{36}$	$\frac{3}{36}$	$\frac{4}{36}$	$\frac{5}{36}$	$\frac{6}{36}$	$\frac{5}{36}$	$\frac{4}{36}$	$\frac{3}{36}$	$\frac{2}{36}$	$\frac{1}{36}$

The mean is

$$E(X) = E\left(\frac{D_1 + D_2}{2}\right).$$

By linearity of expectation,

$$E(X) = \frac{1}{2} [E(D_1) + E(D_2)].$$

For one balanced die,

$$E(D_1) = E(D_2) = \frac{1 + 2 + 3 + 4 + 5 + 6}{6} = 3.5.$$

Therefore,

$$E(X) = \frac{1}{2}(3.5 + 3.5) = 3.5.$$

So,

$$\boxed{\mu_X = 3.5}.$$

For one balanced die,

$$\text{Var}(D_1) = \frac{35}{12}.$$

Since the two rolls are independent,

$$\text{Var}(X) = \text{Var}\left(\frac{D_1 + D_2}{2}\right) = \frac{1}{4} [\text{Var}(D_1) + \text{Var}(D_2)].$$

Thus,

$$\text{Var}(X) = \frac{1}{4} \left(\frac{35}{12} + \frac{35}{12} \right) = \frac{35}{24}.$$

Therefore, the standard deviation is

$$\boxed{\sigma_X = \sqrt{\frac{35}{24}} \approx 1.2076}.$$

Problem 19

Question. Let X and Y denote lengths of life in hundreds of hours for type-1 and type-2 components. The joint density is

$$f(x, y) = \frac{1}{8}xe^{-(x+y)/2}, \quad x > 0, y > 0.$$

Find the marginal densities of X and Y . Determine whether X and Y are independent. Find the marginal mean and variance for each of X and Y . Find the probability that a type-2 component has life length in excess of 100 hours.

Solution. The joint density is

$$f(x, y) = \frac{1}{8}xe^{-(x+y)/2}, \quad x > 0, y > 0.$$

Marginal density of X .

$$f_X(x) = \int_0^\infty f(x, y) dy.$$

Thus,

$$f_X(x) = \int_0^\infty \frac{1}{8}xe^{-(x+y)/2} dy = \frac{x}{8}e^{-x/2} \int_0^\infty e^{-y/2} dy.$$

Since

$$\int_0^\infty e^{-y/2} dy = 2,$$

this gives

$$f_X(x) = \frac{x}{4}e^{-x/2}, \quad x > 0.$$

Marginal density of Y .

$$f_Y(y) = \int_0^\infty f(x, y) dx.$$

Thus,

$$f_Y(y) = \int_0^\infty \frac{1}{8}xe^{-(x+y)/2} dx = \frac{1}{8}e^{-y/2} \int_0^\infty xe^{-x/2} dx.$$

Since

$$\int_0^\infty xe^{-x/2} dx = 4,$$

this gives

$$f_Y(y) = \frac{1}{2}e^{-y/2}, \quad y > 0.$$

Independence. Now,

$$f_X(x)f_Y(y) = \left(\frac{x}{4}e^{-x/2}\right)\left(\frac{1}{2}e^{-y/2}\right) = \frac{x}{8}e^{-(x+y)/2} = f(x, y).$$

Therefore,

$$X \text{ and } Y \text{ are independent.}$$

Mean and variance of X . The marginal density of X is

$$f_X(x) = \frac{x}{4}e^{-x/2}, \quad x > 0.$$

This is a Gamma distribution with $\alpha = 2$ and $\beta = 2$. Therefore,

$$E(X) = \alpha\beta = (2)(2) = 4,$$

and

$$\text{Var}(X) = \alpha\beta^2 = (2)(2^2) = 8.$$

Thus,

$$\boxed{E(X) = 4, \quad \text{Var}(X) = 8}.$$

Mean and variance of Y . The marginal density of Y is

$$f_Y(y) = \frac{1}{2}e^{-y/2}, \quad y > 0.$$

This is an Exponential distribution with rate $\lambda = 1/2$. Therefore,

$$E(Y) = \frac{1}{\lambda} = 2,$$

and

$$\text{Var}(Y) = \frac{1}{\lambda^2} = 4.$$

Thus,

$$\boxed{E(Y) = 2, \quad \text{Var}(Y) = 4}.$$

Since Y is measured in hundreds of hours, a lifetime in excess of 100 hours means

$$Y > 1.$$

Thus,

$$P(Y > 1) = \int_1^\infty \frac{1}{2}e^{-y/2} dy = e^{-1/2}.$$

Therefore,

$$\boxed{P(Y > 1) = e^{-1/2} \approx 0.6065}.$$

Problem 20

Question. The seizure counts for 20 patients are

$$5, 0, 9, 6, 0, 0, 5, 0, 6, 1, 5, 0, 0, 0, 0, 7, 0, 0, 4, 7.$$

Find (a) the mean number of seizures, (b) the five-number summary, and (c) a modified boxplot with fences and outlier detection.

Solution.

(a) The sample mean is

$$\bar{x} = \frac{\sum x_i}{n}.$$

Here,

$$\sum x_i = 55, \quad n = 20.$$

Thus,

$$\boxed{\bar{x} = \frac{55}{20} = 2.75}.$$

(b) Arrange the data in increasing order:

$$0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 1, 4, 5, 5, 5, 6, 6, 7, 7, 9.$$

Since $n = 20$, the median is the average of the 10th and 11th observations:

$$\tilde{x} = \frac{0 + 1}{2} = 0.5.$$

The lower half is

$$0, 0, 0, 0, 0, 0, 0, 0, 0, 0,$$

so

$$Q_1 = 0.$$

The upper half is

$$1, 4, 5, 5, 5, 6, 6, 7, 7, 9,$$

so

$$Q_3 = \frac{5 + 6}{2} = 5.5.$$

Therefore, the five-number summary is

$$\boxed{(0, 0, 0.5, 5.5, 9)}.$$

(c) The interquartile range is

$$IQR = Q_3 - Q_1 = 5.5 - 0 = 5.5.$$

The lower fence is

$$Q_1 - 1.5(IQR) = 0 - 1.5(5.5) = -8.25.$$

The upper fence is

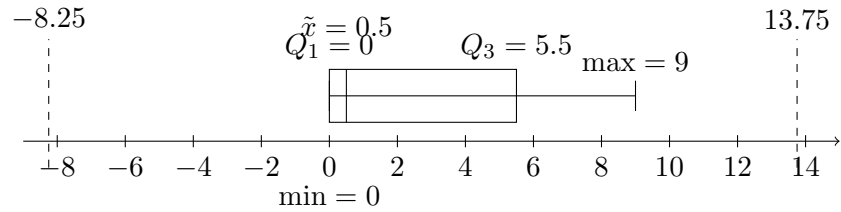
$$Q_3 + 1.5(IQR) = 5.5 + 1.5(5.5) = 13.75.$$

Thus, the fences are

$$-8.25 \text{ and } 13.75.$$

Since all observations are between -8.25 and 13.75 , there are no outliers. The whiskers go from 0 to 9.

No outliers are detected.



Problem 21

Question. To compare Glipizide and Jardiance with a placebo control group, 18 diabetics were randomly assigned into three groups of 6. The morning blood-sugar readings are shown below. Test at a 1% significance level whether all three population mean blood-sugar levels are the same.

	Placebo	Glipizide	Jardiance
1	137	128	119
2	161	139	112
3	155	121	123
4	179	146	108
5	144	151	128
6	150	125	132

Solution. Test whether the three population mean blood-sugar levels are the same.

$$H_0 : \mu_1 = \mu_2 = \mu_3$$

versus

$$H_a : \text{at least one population mean is different.}$$

The sample means are

$$\bar{x}_1 = \frac{137 + 161 + 155 + 179 + 144 + 150}{6} = 154.3333,$$

$$\bar{x}_2 = \frac{128 + 139 + 121 + 146 + 151 + 125}{6} = 135,$$

and

$$\bar{x}_3 = \frac{119 + 112 + 123 + 108 + 128 + 132}{6} = 120.3333.$$

The grand mean is

$$\bar{x}_{..} = \frac{926 + 810 + 722}{18} = 136.5556.$$

The treatment sum of squares is

$$SSTr = \sum_{i=1}^3 n_i (\bar{x}_i - \bar{x}_{..})^2.$$

Thus,

$$SSTr = 6(154.3333 - 136.5556)^2 + 6(135 - 136.5556)^2 + 6(120.3333 - 136.5556)^2.$$

So,

$$SSTr = 3489.7778.$$

The error sum of squares is

$$SSE = \sum_{i=1}^3 \sum_{j=1}^6 (x_{ij} - \bar{x}_i)^2.$$

Computing within each group,

$$SSE = 2242.6667.$$

The degrees of freedom are

$$df_{Tr} = k - 1 = 3 - 1 = 2,$$

and

$$df_E = N - k = 18 - 3 = 15.$$

Therefore,

$$MSTr = \frac{SSTr}{df_{Tr}} = \frac{3489.7778}{2} = 1744.8889,$$

and

$$MSE = \frac{SSE}{df_E} = \frac{2242.6667}{15} = 149.5111.$$

The ANOVA test statistic is

$$F = \frac{MSTr}{MSE} = \frac{1744.8889}{149.5111} = 11.6706.$$

The ANOVA table is

Source	df	SS	MS	F
Treatment	2	3489.7778	1744.8889	11.6706
Error	15	2242.6667	149.5111	
Total	17	5732.4444		

At the 1% significance level,

$$F_{0.01,2,15} = 6.3589.$$

The rejection rule is

$$\text{reject } H_0 \text{ if } F > 6.3589.$$

Since

$$11.6706 > 6.3589,$$

reject H_0 . Therefore, at the 1% significance level, there is sufficient evidence to conclude that not all three population mean blood-sugar levels are the same.

Reject H_0 .
